



Welspun Corporation

SAP Integration for Pipe Testing Stations

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Faclon Labs at a Glance

Leaders in Industrial Data Integration, Visualization, and AI based Optimization



183+
Customers

400+
Deployments

60k+
Assets
Connected

5 B+
Data Points
Daily

150+
Employees



Rishi Sharma
CEO
B.Tech IIT Bombay



Utkarsh Srivastava
CTO
Ex-White Hat Hacker



Parth Chaudhary
President of Solution
B. Tech IIT Bombay



Vivek Singh
Chief Data Scientist
B.Tech IIT Kanpur



Archit Naraniwal
Chief Sales Officer
B. Tech IIT Bombay



Parag Patil
Chief Strategic Alliances
Ex-Wipro, Infosys, RIL

Backed By



Sarjan Shah
MD, Group Satellite



Jay Kotak
Co-head, Kotak 811



Siddharth Mehra
ED, Savita Oil

Key Customers

Trusted and Ideal digital partner for Industrial setups, commercial setups and OEMs

Automobile & ancillary



TATA HITACHI



Consumer Goods



Metals



HINDUSTAN ZINC



Cement



Pharma



Consumer Durables



Commercial Infra



Energy, Oil & Gas



OEM tie-ups



Partnerships

accenture



Objective & Business Outcome

Automatically capture each test result, let the operator review it, and push it to SAP with the correct rules applied.

🕒 CURRENT PROCEDURE



01

UT Stations

IUT & FUT · Ultrasonic testing

APPLICATION

SAP Entry Application

- 👁️ Operator **counts the indications** shown on the UT machine
- 📄 Details **typed by hand** into SAP against each pipe number

⌛ Slow

⚠️ Prone to mistakes

🎯 OBJECTIVE



Capture each result automatically from the machine's PDF report and JPEG image



Decide the result from the prove-up rules saved per PO: MUT, X-Ray or SAT



Operator reviews and uploads to SAP with the right endpoint and timestamp



✅ BUSINESS OUTCOME



Faster entry: no manual counting or re-typing



Fewer mistakes: values parsed from the machine report



Consistent results: PO rules applied to every pipe



Traceable records: PDF report and JPEG image kept with each entry



02

Hydrotester

Hydrostatic pressure testing

APPLICATIONS

Graph Application + SAP Entry Application

- 📄 Test results **entered directly into SAP** by hand
- 📄 Pressure graph produced on a **manual printer**

⌛ Slow

⚠️ Prone to mistakes



Pipe number entered before every test — scanned, selected or typed



Capture the pressure-vs-time curve digitally from the SCADA over OPC UA



Operator reviews, corrects and uploads to SAP with both graph images



Printer eliminated: graphs captured and stored digitally



Fewer mistakes: values captured from SCADA; corrections kept on record



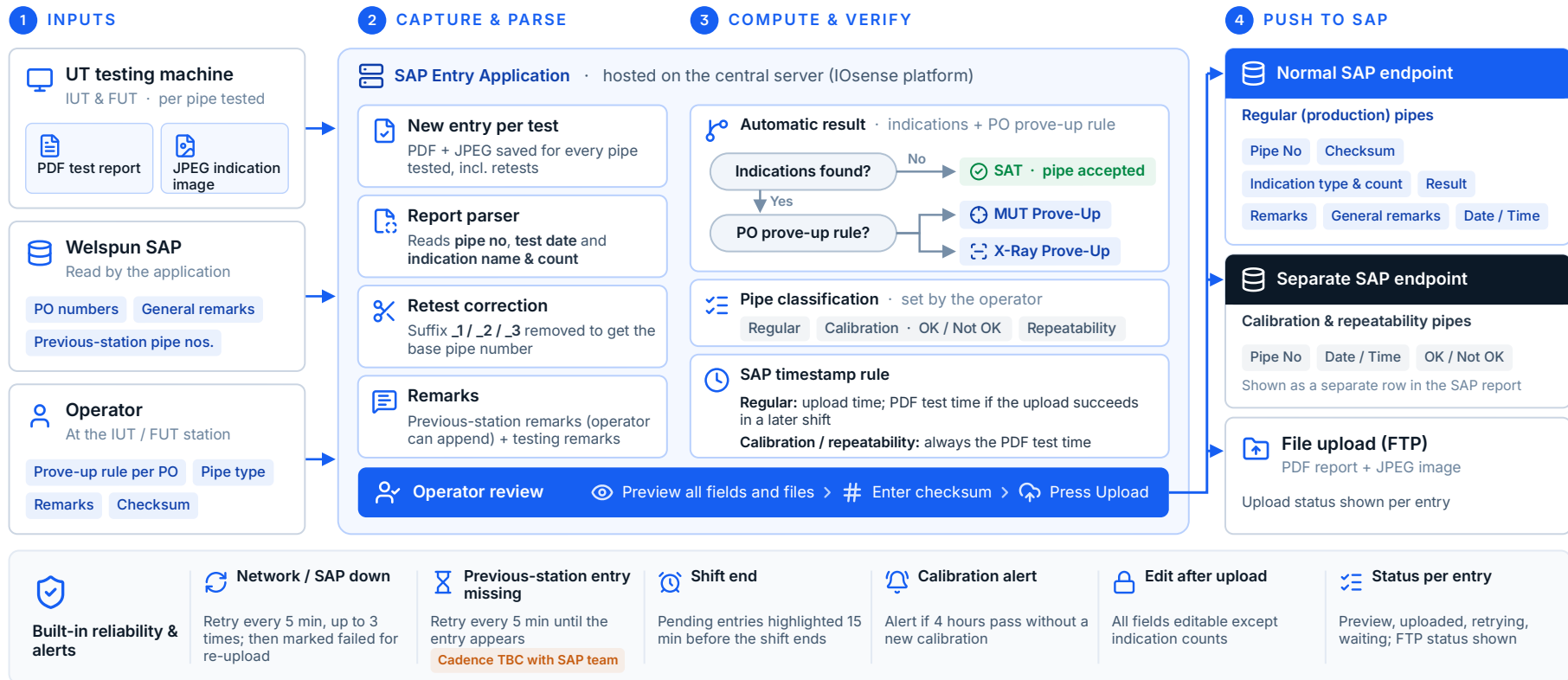
Right pipe, right record: no pipe number, no saved test



Complete records: standard and actual graphs with each entry

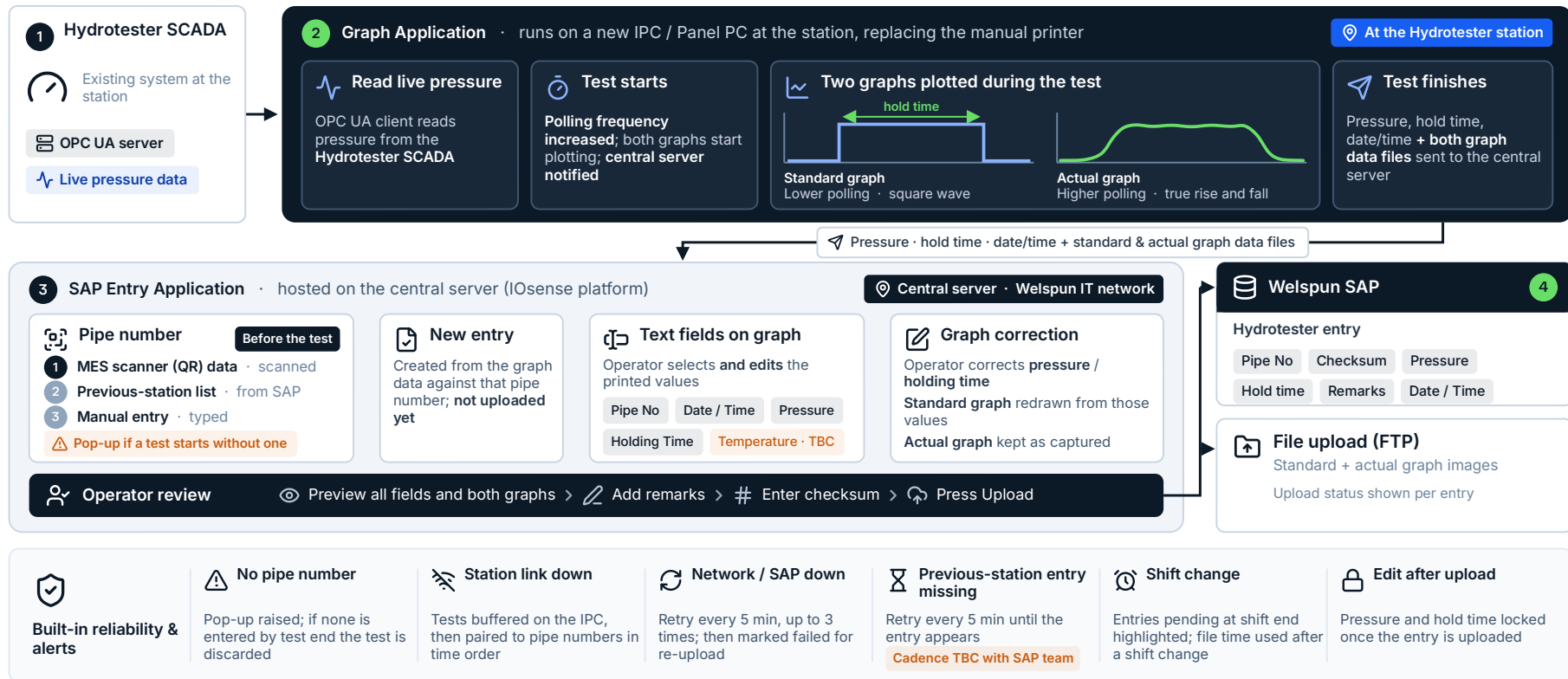
Solution Architecture

What comes in, what the SAP Entry Application computes and verifies, and what reaches SAP.



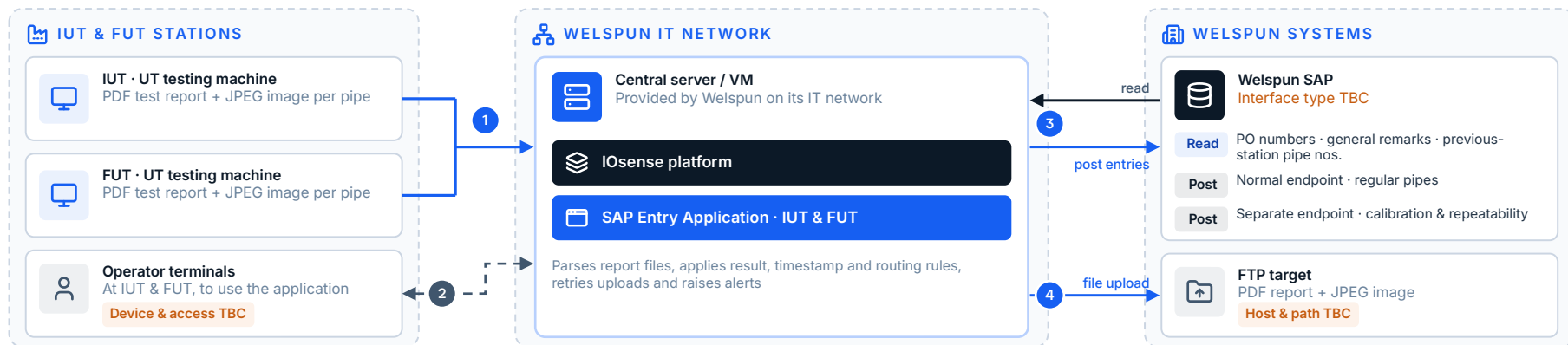
Solution Architecture

The pipe number is entered before the test; the Graph Application captures the curve and the SAP Entry Application posts the entry.



Network Architecture

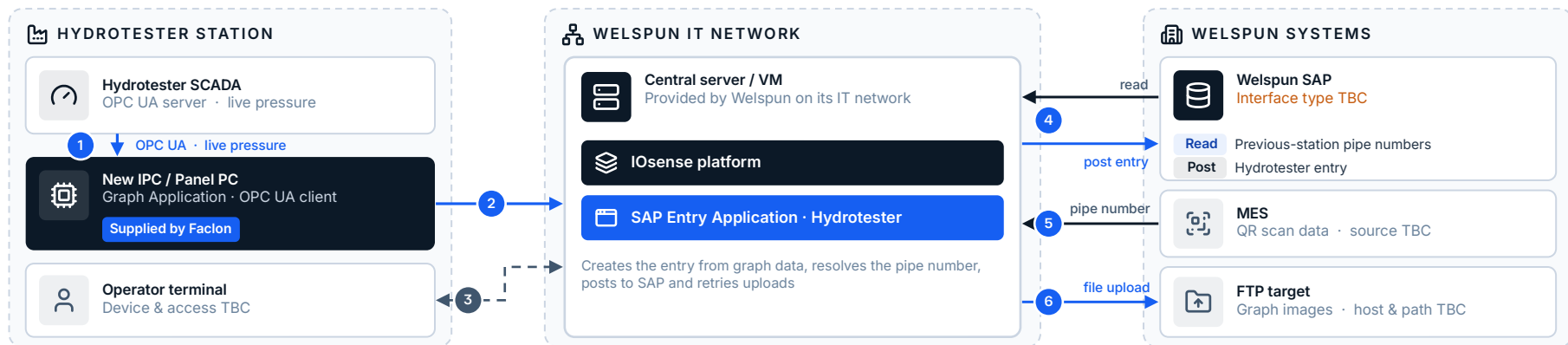
Data flow and configuration for the UT stations. Items marked TBC are to be confirmed with Welspun IT and SAP teams.



REF	LINK / COMPONENT	WHAT FLOWS	CONFIGURATION TO SET UP	STATUS
1	IUT & FUT machines → server	PDF test report + JPEG indication image for every pipe tested	Transfer method, file location and trigger	🔍 TBC
2	Operator terminals ↔ application	Review, classify pipe, enter checksum, upload	Terminal device and access method at IUT & FUT	🔍 TBC
3	Central server ↔ SAP	Read: PO numbers, general remarks, previous-station pipe nos. · Post: normal and separate endpoints	SAP interface type, endpoint details, credentials	🔍 TBC
4	Central server → FTP	PDF report + JPEG image; upload status shown per entry	FTP host, folder and credentials	🔍 TBC
●	Central server / VM	Hosts the IOsense platform and the SAP Entry Application	On Welspun IT network, provided by Welspun	✅ Defined
●	Upload retry	Network / SAP down · previous-station entry missing	Every 5 min, up to 3 tries · every 5 min until available	🔍 Cadence TBC
●	Network access	Stations, central server, SAP and FTP	Ports, VLANs and firewall rules	🔍 TBC

Network Architecture

Data flow and configuration for the Hydrotester station. Items marked TBC are to be confirmed with Welspun IT and SAP teams.



REF	LINK / COMPONENT	WHAT FLOWS	CONFIGURATION TO SET UP	STATUS
1	SCADA ↔ IPC (OPC UA)	Live pressure; polling frequency increased when a test starts	Endpoint, security, pressure tag, polling rates	🔍 TBC
2	IPC → central server	Test-start signal; pressure, hold time, date/time + graph files	Transfer protocol, routing, local buffering	🔍 TBC
3	Operator terminal ↔ application	Pipe number before the test, edits, corrections, upload	Terminal device and access method	🔍 TBC
4	Central server ↔ SAP	Read previous-station pipe numbers · post the Hydrotester entry	SAP interface type, endpoint details, credentials	🔍 TBC
5	MES → central server	QR scan data fills the pipe number before the test	MES data source and interface	🔍 TBC
6	Central server → FTP	Standard + actual graph images; upload status per entry	FTP host, folder and credentials	🔍 TBC
●	Hosting	New IPC runs the Graph Application · central server runs the SAP Entry Application	IPC supplied by Faclox · server / VM provided by Welspun	✅ Defined
●	Network access	Station, central server, SAP, MES and FTP	Ports, VLANs and firewall rules	🔍 TBC

Scope of Work

What Faclon Labs delivers and what Welspun provides for the UT stations project.

 **Faclon Labs scope** design, build, integrate, deploy and support



SAP Entry Application for IUT & FUT · design and development

- **PO configuration:** PO list from SAP; MUT and / or X-Ray prove-up rule saved per PO
- **Report capture:** PDF + JPEG saved; pipe no, test date, indication name & count parsed
- **Result logic:** MUT Prove-Up, X-Ray Prove-Up or SAT; pipe classification; retest suffix removal
- **Upload:** preview and checksum; normal or separate SAP endpoint with timestamp rules; FTP file upload
- **Reliability:** retries, shift-end highlight, 4-hour calibration alert, edit rules, status, previous-station tab



Integration and deployment

- Integration with Welspun SAP endpoints (read and post) and the FTP target
- Deployment on the Welspun-provided central server as part of the IOSense platform



Services

- User manual / SOP · operator training · UAT and go-live support · post go-live support



Welspun scope inputs, access and infrastructure



Central server / VM on the Welspun IT network to host the IOSense platform



SAP endpoints to read PO numbers, general remarks and previous-station pipe numbers



SAP endpoints to post regular and calibration / repeatability entries



SAP team support, incl. confirming the retry cadence for missing previous-station entries



Access to IUT & FUT machine report files (PDF test report + JPEG indication image)



FTP target and credentials for file uploads



Network connectivity between stations, central server, SAP and FTP



Operator and SAP team availability for UAT and training



Travel and accommodation for Faclon engineers during site visits

 **Responsibility to be agreed** Operator terminals at the IUT & FUT stations

Scope of Work

What Faclon Labs delivers and what Welspun provides for the Hydrotester project.



Faclon Labs scope design, build, integrate, deploy and support



Hardware and Graph Application on the IPC · design and development

- Supply of a new **IPC / Panel PC** for the Hydrotester station
- **OPC UA connection** to the Hydrotester SCADA; polling frequency increased when a test starts
- **Standard graph** (square wave, hold time) and **actual graph** (true rise and fall of pressure)
- **At test finish**: pressure, hold time, date/time and both graph data files pushed to the central server
- **If the link is down**: tests buffered on the IPC and forwarded automatically when it returns



SAP Entry Application for the Hydrotester · design and development

- **Pipe number before every test**: scan, previous-station list or manual entry; pop-up if a test starts without one; test discarded if none entered by test end
- **Entry creation** from graph data; tests and pipe numbers paired in time order after a connectivity loss
- **Upload**: text fields selected and edited; pressure / holding time corrected (standard graph redrawn, actual graph kept as captured); preview, remarks, checksum; FTP upload
- **Reliability**: retries, shift-end highlight, edit rules (pressure & hold time locked after upload), status, previous-station tab



Integration, deployment and services

- Integration with SAP, MES data and the FTP target; deployment on the central server; set-up on the IPC
- User manual / SOP · operator training · UAT and go-live support · post go-live support



Welspun scope inputs, access and infrastructure



Central server / VM on the Welspun IT network to host the IOSense platform



Access to the Hydrotester SCADA OPC UA server: endpoint, credentials and pressure tag details



Access to MES scanner (QR) data for pipe numbers



SAP endpoints to read previous-station pipe numbers and post Hydrotester entries



SAP team support, incl. confirming the retry cadence for missing previous-station entries



FTP target and credentials for graph image uploads



Network connectivity: IPC to SCADA, IPC to central server, server to SAP, MES and FTP



Operator and SAP team availability for UAT and training



Travel and accommodation for Faclon engineers during site visits

⌚ **Responsibility to be agreed** Operator terminal at the Hydrotester station · IPC mounting, power and cabling at the station

Software Deliverables

One application for the IUT and FUT stations, delivered as six modules.



SAP Entry Application · IUT & FUT

 Central server IOsense platform

PO Configuration

01

- PO dropdown filled automatically from SAP
- MUT and / or X-Ray prove-up rule set per PO
- Settings reused for every pipe on that PO



Report Capture & Parser

02

- New entry for every pipe tested, incl. retests
- PDF test report and JPEG indication image saved
- Pipe no, test date, indication name & count parsed



Result & Classification

03

- Auto result: MUT Prove-Up, X-Ray Prove-Up or SAT
- Regular, calibration (OK / Not OK) or repeatability
- Retest suffix (_1, _2, _3) removed from pipe no



Remarks & Review

04

- General remarks from the previous station, with append
- Testing remarks for this station
- Preview of all fields and files; checksum input



SAP Upload

05

- Regular pipes posted to the normal SAP endpoint
- Calibration / repeatability to the separate endpoint
- Timestamp rules for shift change; FTP file upload



Reliability & Monitoring

06

- Retries on failure; waits for previous-station entry
- 15-min shift-end highlight; 4-hour calibration alert
- Edit rules, entry and FTP status, previous-station tab



Services
included in scope



User manual / SOP
Documentation for operators and administrators



Operator training
For station operators and supervisors



UAT & go-live support
Through user acceptance testing and go-live



Post go-live support
Support after go-live
⌚ Duration TBC

Software Deliverables

Two applications for the Hydrotester station: a Graph Application on the new IPC and a SAP Entry Application on the central server.



Graph Application

 Station IPC



SAP Entry Application · Hydrotester

 Central server

 IOsense platform



OPC UA Connector

01

- Reads live pressure from the Hydrotester SCADA
- Polling frequency increased when a test starts



Entry Creation

04

- New entry from graph data for every test
- Held until the operator uploads
- After an outage, tests and pipe numbers paired in time order



Pipe Number Capture

05

- **Before every test:** MES scan, previous-station list or manual
- Pop-up if a test starts without one
- Test discarded if none entered by test end



Graph Capture

02

- **Standard graph:** lower polling, square wave, hold time
- **Actual graph:** higher polling, true rise and fall



Graph Image & Correction

06

- Selects and edits the printed text fields
- Corrects pressure / holding time
- Standard graph redrawn; actual kept as captured



SAP Upload

07

- Remarks and checksum, then Upload to SAP
- Upload time; test time from the file after a shift change
- FTP upload of standard and actual graph images



Result Publisher

03

- At test finish: pressure, hold time, date/time
- Both graph data files pushed to the central server
- Buffered on the IPC if the link is down



Reliability & Alerts

08

- Retry every 5 min, up to 3 times, on network / SAP failure
- Retry every 5 min while previous-station entry is missing
- Pending entries highlighted 15 min before shift end



Entry Management

09

- Edit uploaded entries (pressure & hold time locked)
- Entry status and FTP upload status
- Previous-station tab with pipe numbers and remarks



Services
included in scope



User manual / SOP
Documentation for operators and administrators



Operator training
For station operators and supervisors



UAT & go-live support
Through user acceptance testing and go-live



Post go-live support
Support after go-live
⚠️ Duration TBC

Implementation Milestones

The sequence each project follows from kick-off to go-live; the week-by-week plan follows on the next slide.

01



UT Stations

SAP Entry Application



MILESTONE 1

Kick-off & requirement sign-off

Scope, SAP data points and business rules signed off



MILESTONE 2

Development

PO configuration, report parser, result logic, upload and alerts



MILESTONE 3

Integration

SAP read and post endpoints, FTP upload, deployment on the central server



MILESTONE 4

UAT

End-to-end testing with operators at the IUT and FUT stations



MILESTONE 5

Go-live & handover

Go-live, user manual / SOP and operator training; post go-live support begins

02



Hydrotester

Graph Application + SAP Entry Application



MILESTONE 1

Kick-off & requirement sign-off

Scope, SAP data points and graph text fields signed off



MILESTONE 2

Development

Graph Application and SAP Entry Application for the Hydrotester



MILESTONE 3

Integration

IPC set-up at the station, OPC UA link to SCADA, MES data, SAP and FTP



MILESTONE 4

UAT

End-to-end testing with operators at the Hydrotester station



MILESTONE 5

Go-live & handover

Go-live, user manual / SOP and operator training; post go-live support begins

Project Delivery Schedule

Indicative plan from kick-off. Hardware supply runs alongside development; SAP integration, UAT and training follow the SAP endpoints being available.



Commercials

Commercial summary for the UT stations project (IUT & FUT).

#	LINE ITEM	DESCRIPTION	TYPE	AMOUNT
01	Software development & implementation	SAP Entry Application for the IUT & FUT stations: development, SAP integration and deployment	One-time	
02	IOsense platform licence	Platform licence for the central server deployment	Annual	
03	Annual support / AMC	Annual maintenance and support	Annual	



One-time total

Software development & implementation



Annual recurring total

IOsense platform licence + Annual support / AMC



Commercials

Commercial summary for the Hydrotester project.

#	LINE ITEM	DESCRIPTION	TYPE	AMOUNT
01	Software development & implementation	Graph Application and SAP Entry Application for the Hydrotester: development, OPC UA and SAP integration, deployment	One-time	
02	IPC / Panel PC hardware	New IPC / Panel PC for the Hydrotester station	One-time	
03	IOsense platform licence	Platform licence for the central server deployment	Annual	
04	Annual support / AMC	Annual maintenance and support	Annual	



One-time total

Software development & implementation + IPC / Panel PC hardware



Annual recurring total

IOsense platform licence + Annual support / AMC

Open Points to Confirm

Items to close with Welspun before requirement sign-off. Tags show the project each point applies to.



SAP & business rules

- 01 Retry cadence while the previous-station entry is missing (SAP team) 01 02
- 02 SAP interface type and endpoint details for reads and posts 01 02
- 03 Shift timings for the shift-end highlight and shift-change timestamp rule 01 02
- 04 Fifth graph text field — Temperature in the task list, TBC in §3.2 02
- 05 Scope-doc update: pressure & hold time correctable before upload 02



Integration & data sources

- 06 Transfer method for IUT / FUT report files (PDF + JPEG) to the central server 01
- 07 OPC UA server details: endpoint, security and pressure tag; how test start and finish are identified 02
- 08 Polling rates for the standard and actual graphs 02
- 09 MES scanner (QR) data source and interface 02



Network & site

- 10 FTP host, folder and credentials for file uploads 01 02
- 11 Ports, VLANs and firewall rules between stations, central server, SAP, MES and FTP 01 02
- 12 Operator terminals at IUT, FUT and Hydrotester: device and access method 01 02
- 13 IPC mounting, power and cabling at the Hydrotester station 02



Plan & support

- 14 Post go-live support duration 01 02
- 15 Confirmation of the delivery schedule and phase durations 01 02